# HTML NOTES
# For Beginners

Theory • Important Tags • Attributes • Code Examples • Practice Tasks

Topics covered

HTML introduction and structure, headings, paragraphs and text formatting, images, hyperlinks, lists, tables, forms, block and inline elements, div and span.

Reference playlist: HTML Tutorial Playlist on YouTube

Beginner-friendly learning notes

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How to use these notes: Type each example into an index.html file, save it, and open it in a browser. Change the content and attributes to see what happens.

# 1. Introduction to HTML

## What is HTML?

HTML stands for HyperText Markup Language. It is the standard markup language used to create and structure web pages. HTML tells the browser what content is present and how it is organized, including headings, paragraphs, images, links, tables, and forms.

HTML is a markup language rather than a general-purpose programming language because it does not provide programming logic by itself.

## HyperText and Markup Language

HyperText is text containing links that can take users to other documents or pages. A markup language uses tags to describe and organize content.

```
<h1>Welcome to HTML</h1>
<p>This is my first web page.</p>
```

## Tag, element, and attribute

A tag is a keyword enclosed in angle brackets. An element usually consists of an opening tag, content, and a closing tag. An attribute provides additional information and is usually written in the opening tag.

```
<p class="intro">Hello World</p>
```

-  is the opening tag.
- Hello World is the content.
-  is the closing tag.
- class="intro" is an attribute and its value.

Some elements, such as img and br, are void elements and do not have closing tags.

## Common attributes

| Attribute | Purpose |
|---|---|
| id | Gives an element a unique identifier within the document. |
| class | Groups elements for CSS styling or scripting. |
| style | Applies inline CSS. |
| title | Provides advisory information, often shown as a tooltip. |
| href | Specifies a link destination. |
| src | Specifies the source of an image or other resource. |
| alt | Provides alternative text for an image. |
| width / height | Specifies dimensions where supported. |

```
<p id="intro" class="highlight" title="Introduction">
    Welcome to my website.
</p>
```

# 2. Basic Structure of an HTML Document

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My First Website</title>
</head>
<body>
    <h1>Welcome to My Website</h1>
    <p>This is my first HTML page.</p>
</body>
</html>
```

| Tag or declaration | Explanation |
|---|---|
| <!DOCTYPE html> | Declares an HTML5 document. |
| <html> | The root element of the document. |
| <head> | Contains metadata and resource information. |
| <meta charset="UTF-8"> | Sets the character encoding. |
| <meta name="viewport"> | Helps the page adapt to different screen sizes. |
| <title> | Sets the browser tab title. |
| <body> | Contains the page content shown in the browser. |

## Head versus body

head: Contains information about the document, such as its title, metadata, and links to CSS files.
body: Contains the page content, such as headings, paragraphs, images, links, tables, and forms.

## HTML comments

Comments are notes inside the source code. They are not displayed as normal page content.

```
<!-- This is an HTML comment -->
<h1>HTML Tutorial</h1>
```

# 3. HTML Headings

HTML provides six heading levels, from h1 to h6. Use them to organize content into sections and subsections.

| Tag | Purpose |
|---|---|
| <h1> | Highest-level heading |
| <h2> | Second-level heading |
| <h3> | Third-level heading |
| <h4> | Fourth-level heading |
| <h5> | Fifth-level heading |
| <h6> | Sixth-level heading |

```
<h1>Web Development</h1>
<h2>HTML</h2>
<h3>HTML Introduction</h3>
<h4>HTML Elements</h4>
<h5>HTML Tags</h5>
```

```
<h6>HTML Attributes</h6>
```

- Use headings according to the logical structure of the page.
- Do not choose a heading only to make text larger or smaller.
- Use CSS to customize font size, color, and appearance.

# 4. Paragraphs and Text Formatting

## Paragraph, line break, and thematic break

| Tag | Purpose |
|---|---|
| <p> | Defines a paragraph. |
| <br> | Inserts a line break; it is a void element. |
| <hr> | Represents a thematic break, often shown as a horizontal line. |
| <pre> | Preserves spaces and line breaks in text. |

```
<p>HTML is used to structure web pages.</p>
<p>CSS is used to style web pages.</p>

<p>
    Ankur Singh<br>
    Web Development Trainer<br>
    Jaipur, Rajasthan
</p>

<h2>Introduction</h2>
<p>This section introduces HTML.</p>
<hr>
<h2>Conclusion</h2>
<p>This section summarizes HTML.</p>
```

## Text formatting tags

| Tag | Purpose | Example |
|---|---|---|
|  | Draws attention with bold styling. | Important |
|  | Indicates strong importance. | Warning |
|  | Alternate voice, mood, or convention. | Term |
|  | Indicates emphasis. | Read carefully |
| <u> | Marks text with an underline. | <u>Underlined</u> |
| <mark> | Highlights text. | <mark>Important</mark> |
| <small> | Small print or side comments. | <small>Terms apply</small> |
| <del> | Represents deleted content. | <del>₹1000</del> |
| <ins> | Represents inserted content. | <ins>₹800</ins> |
| <sub> | Displays subscript text. | H<sub>2</sub>O |
| <sup> | Displays superscript text. | x<sup>2</sup> |

```
<p><b>This text is bold.</b></p>
<p><strong>This text is important.</strong></p>
<p><i>This text is italic.</i></p>
<p><em>This text is emphasized.</em></p>
<p><mark>This text is highlighted.</mark></p>
<p>Water formula: H<sub>2</sub>O</p>
<p>Square of x: x<sup>2</sup></p>
```

```
<p>Old price: <del>■1000</del> New price: <ins>■800</ins></p>

<pre>
Name: Ankur
Course: HTML
Duration: 30 Days
</pre>
```

Remember: b and i are mainly presentational/conventional, while strong and em convey importance and emphasis. Use semantic meaning where appropriate.

# 5. Images in HTML

The img tag displays an image. It is a void element and does not require a closing tag.

```
<img src="image.jpg" alt="Description of image">
```

| Attribute | Purpose |
|---|---|
| src | Specifies the image location. |
| alt | Provides alternative text for the image. |
| width / height | Specifies dimensions. |
| title | Provides advisory information. |
| loading | Can control when the browser loads an image. |

```
<img
    src="student.jpg"
    alt="A student studying"
    width="300"
    height="200"
>
```

## Absolute and relative paths

An absolute URL gives the full web address of an image. A relative path locates an image in relation to the current HTML file.

```
<!-- Absolute URL (illustrative domain) -->
<img src="https://example.com/images/student.jpg"
     alt="Student studying" width="300">

<!-- Relative path: image inside an images folder -->
<img src="images/student.jpg" alt="Student studying">

<!-- Image in the same folder as index.html -->
<img src="student.jpg" alt="Student studying">

<!-- Image in the parent folder -->
<img src="../student.jpg" alt="Student studying">
```

Example project structure:

```
my-website/
■■■ index.html
■■■ images/
    ■■■ student.jpg
```

## Clickable image

```
<a href="https://example.com">
    <img src="images/logo.png" alt="Visit our website">
</a>
```

If an image does not appear, check the file path, filename, extension, spelling, and whether the image file is available. Use meaningful alt text for informative images and alt="" for purely decorative images.

# 6. Anchor Tag and Hyperlinks

The a tag creates a hyperlink to another page, website, file, email address, or section of a page.

```
<a href="destination">Link Text</a>
```

| Attribute | Purpose |
|---|---|
| href | Specifies the destination. |
| target | Specifies where to open the destination. |
| title | Provides advisory information. |
| download | Suggests downloading a resource. |
| rel | Specifies the relationship to the linked document. |

## External link and new tab

```
<a href="https://www.wikipedia.org">Visit Wikipedia</a>

<a href="https://www.wikipedia.org"
   target="_blank"
   rel="noopener noreferrer">
    Open Wikipedia
</a>
```

## Navigation between local pages

```
website/
■■■ index.html
■■■ about.html

<!-- Inside index.html -->
<a href="about.html">About Us</a>
```

## In-page navigation

```
<a href="#contact">Go to Contact Section</a>

<h2>About Us</h2>
<p>Information about our website.</p>

<h2 id="contact">Contact Us</h2>
<p>Email us for more information.</p>
```

## Email and download links

```
<a href="mailto:info@example.com">Send Email</a>

<a href="files/notes.pdf" download>Download Notes</a>
```

The mailto link generally opens the user's configured email application. The download attribute is subject to browser and resource-origin restrictions.

# 7. Lists in HTML

Lists organize related items. The main types are ordered lists, unordered lists, and description lists.

## Ordered list: ol

An ordered list shows items in a sequence, commonly numbered. The li tag defines each item.

```
<h2>Steps to Create a Website</h2>
<ol>
    <li>Learn HTML</li>
    <li>Learn CSS</li>
    <li>Learn JavaScript</li>
    <li>Build a Project</li>
</ol>
```

| Attribute | Purpose |
| --- | --- |
| type | Specifies numbering style, such as 1, A, a, I, or i. |
| start | Specifies the starting number. |
| reversed | Reverses the numbering order. |

```
<ol type="A" start="3">
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
</ol>
```

This list begins with C, then D and E.

## Unordered list: ul

```
<h2>Web Technologies</h2>
<ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
</ul>
```

## Nested list

```
<ul>
    <li>Frontend
        <ul>
            <li>HTML</li>
            <li>CSS</li>
            <li>JavaScript</li>
        </ul>
    </li>
    <li>Backend
        <ul>
            <li>Node.js</li>
            <li>Express.js</li>
        </ul>
    </li>
</ul>
```

## Description list: dl

```
<dl>
    <dt>HTML</dt>
    <dd>Used to structure web pages.</dd>

    <dt>CSS</dt>
    <dd>Used to style web pages.</dd>

    <dt>JavaScript</dt>
    <dd>Used to add behavior to web pages.</dd>
</dl>
```

dl defines the description list, dt defines a term, and dd provides its description.

# 8. Tables in HTML

An HTML table displays data in rows and columns. Tables are useful for student records, timetables, marks, and other tabular data.

| Tag | Purpose |
|---|---|
| <table> | Defines a table. |
| <caption> | Provides a table title. |
| <tr> | Defines a table row. |
| <th> | Defines a header cell. |
| <td> | Defines a data cell. |
| <thead> | Groups header rows. |
| <tbody> | Groups the main data rows. |
| <tfoot> | Groups footer rows. |

## Basic table

```
<table border="1">
    <caption>Student Records</caption>
    <tr>
        <th>Roll No.</th>
        <th>Name</th>
        <th>Course</th>
    </tr>
    <tr>
        <td>101</td>
        <td>Aman</td>
        <td>BCA</td>
    </tr>
    <tr>
        <td>102</td>
        <td>Priya</td>
        <td>B.Tech</td>
    </tr>
</table>
```

The border attribute is obsolete in modern HTML. It is included here only to demonstrate the basic structure. Use CSS for borders in real projects.

## Semantic table sections

```
<table>
    <caption>Student Marks</caption>
    <thead>
        <tr><th>Student</th><th>HTML</th><th>CSS</th></tr>
    </thead>
    <tbody>
        <tr><td>Aman</td><td>85</td><td>90</td></tr>
        <tr><td>Priya</td><td>92</td><td>88</td></tr>
    </tbody>
</table>
```

## colspan: span multiple columns

```
<table border="1">
    <tr><th colspan="3">Student Information</th></tr>
    <tr><td>101</td><td>Aman</td><td>BCA</td></tr>
</table>
```

### rowspan: span multiple rows

```
<table border="1">
    <tr><th>Department</th><th>Student</th></tr>
    <tr><td rowspan="2">Computer Science</td><td>Aman</td></tr>
    <tr><td>Priya</td></tr>
</table>
```

| Attribute | Purpose |
|---|---|
| colspan | Makes a cell span multiple columns. |
| rowspan | Makes a cell span multiple rows. |
| scope | Identifies which cells a header applies to. |
| headers | Associates a data cell with header cells by ID. |

```
<table>
    <tr>
        <th scope="col">Name</th>
        <th scope="col">Marks</th>
    </tr>
    <tr><td>Aman</td><td>85</td></tr>
</table>
```

# 9. Forms in HTML

HTML forms collect information from users. Common examples include login, registration, contact, feedback, and admission forms.

| Tag | Purpose |
|---|---|
| <form> | Defines the form. |
| <label> | Provides a label for a control. |
| <input> | Creates an input control. |
| <textarea> | Accepts multiline text. |
| <select> | Creates a dropdown list. |
| <option> | Defines a dropdown option. |
| <button> | Creates a button. |
| <fieldset> | Groups related controls. |
| <legend> | Provides a caption for a fieldset. |

## Important form attributes

| Attribute | Purpose |
| --- | --- |
| action | URL that receives submitted data. |
| method | HTTP method, commonly get or post. |
| name | Identifies a submitted field. |
| type | Specifies the type of input control. |
| placeholder | Displays a hint inside a control. |
| required | Requires a value before submission. |
| value | Specifies a control's value. |
| min / max | Sets numeric/date limits where supported. |
| minlength / maxlength | Sets text-length limits where supported. |
| checked / selected | Sets an initial checkbox/radio or option selection. |
| multiple | Allows multiple selections where supported. |
| readonly / disabled | Controls editing or interaction. |
| autocomplete | Controls browser autofill behavior. |

## Basic registration form

```
<form action="/register" method="post">
    <label for="fullname">Full Name:</label>
    <input type="text" id="fullname" name="fullname" required>
    <br><br>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required>
    <br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required>
    <br><br>

    <button type="submit">Register</button>
</form>
```

This example demonstrates structure. The /register endpoint must exist on a server to process the submission.

## Common input types

| Type | Purpose |
|---|---|
| text | Single-line text |
| password | Password entry |
| email | Email address |
| number | Numeric input |
| tel | Telephone number |
| date / time | Date or time selection |
| datetime-local | Local date and time |
| month / week | Month or week selection |
| checkbox | Independent selections |
| radio | One selection from a group |
| file | File selection |
| color / range | Color picker or range slider |
| url / search | Web address or search text |
| hidden | Hidden form value |
| submit / reset / button | Submit, reset, or general-purpose button |

## Radio buttons

```
<p>Select your gender:</p>
<input type="radio" id="male" name="gender" value="male">
<label for="male">Male</label>

<input type="radio" id="female" name="gender" value="female">
<label for="female">Female</label>

<input type="radio" id="other" name="gender" value="other">
<label for="other">Other</label>
```

Radio buttons with the same name form a group, allowing one option to be selected.

## Checkboxes

```
<p>Select your skills:</p>
<input type="checkbox" id="html" name="skills" value="HTML">
<label for="html">HTML</label>

<input type="checkbox" id="css" name="skills" value="CSS">
<label for="css">CSS</label>

<input type="checkbox" id="js" name="skills" value="JavaScript">
<label for="js">JavaScript</label>
```

Checkboxes allow independent choices.

### Dropdown and textarea

```
<label for="course">Select Course:</label>
<select id="course" name="course">
    <option value="">Choose a course</option>
    <option value="bca">BCA</option>
    <option value="btech">B.Tech</option>
    <option value="mca">MCA</option>
</select>

<label for="message">Your Message:</label>
<textarea id="message" name="message" rows="5" cols="30"
          placeholder="Write your message"></textarea>
```

### Fieldset and legend

```
<form>
    <fieldset>
        <legend>Personal Information</legend>
        <label for="name">Name:</label>
        <input type="text" id="name" name="name">
        <br><br>
        <label for="dob">Date of Birth:</label>
        <input type="date" id="dob" name="dob">
    </fieldset>
</form>
```

### Submit and reset

```
<form>
    <label for="student-name">Name:</label>
    <input type="text" id="student-name" name="name">
    <button type="submit">Submit</button>
    <button type="reset">Reset</button>
</form>
```

### Important form concepts

- A control usually needs a name attribute for its value to be included in form submission.
- Associate labels with controls using matching for and id values.
- required provides basic browser-side validation.
- GET generally places submitted data in the URL query string; POST sends data in the request body.
- HTML validation does not replace server-side validation.
- Sensitive data should be transmitted over HTTPS and handled securely on the server.

# 10. Block-Level and Inline Elements

## Block-level elements

A block-level element generally starts on a new line and occupies the available width of its containing block under normal layout conditions.

Examples include div, p, h1 to h6, ul, ol, li, table, and form.

```
<h1>First Heading</h1>
<p>First paragraph.</p>
<p>Second paragraph.</p>
```

## Inline elements

An inline element generally remains within the current line and occupies only the space required by its content.

Examples include span, a, strong, em, b, i, and img.

```
<p>
    Learn <strong>HTML</strong>,
    <em>CSS</em>, and
    <a href="https://example.com">JavaScript resources</a>.
```

```
</p>
```

Block and inline are default display behaviors, not permanent rules. CSS can change an element's display behavior.

# 11. Difference Between div and span

Both div and span are generic containers. Neither has a specific semantic meaning by itself.

## The div tag

The div element is a generic container commonly used to group larger sections of content. It is block-level by default.

```
<div>
    <h2>About Us</h2>
    <p>We provide web development training.</p>
</div>

<div>
    <h2>Our Courses</h2>
    <p>We teach HTML, CSS, and JavaScript.</p>
</div>
```

## The span tag

The span element is a generic inline container commonly used to target a small part of text or other inline content.

```
<p>
    HTML is a
    <span style="color: blue;">markup language</span>.
</p>
```

| Feature | div | span |
|---|---|---|
| Default display | Block | Inline |
| Typical use | Group sections of content | Group part of inline content |
| Starts a new line by default | Yes | No |
| Semantic meaning | Generic | Generic |
| Can use class and id | Yes | Yes |
| Can be styled with CSS | Yes | Yes |

## Using class and id

```
<div class="card" id="student-card">
    <h2>Student Details</h2>
    <p>Name: Aman</p>
    <p>Course: <span class="course-name">BCA</span></p>
</div>
```

- class can be shared by multiple elements.
- An id should be unique within the document.
- CSS and JavaScript can use these attributes to select elements.

### Changing display with CSS

```
<style>
    div { display: inline; }
    span { display: block; }
</style>

<div>This div is displayed inline.</div>
<span>This span is displayed as a block.</span>
```

# 12. Practice Questions

## A. Theory questions

1. What is HTML? Write its full form.
2. What is the difference between an HTML tag and an HTML element?
3. What are HTML attributes? Give three examples.
4. Explain the basic structure of an HTML document.
5. What is the purpose of head and body?
6. Explain the six HTML heading tags.
7. What is the difference between br and hr?
8. Differentiate between b and strong.
9. What is the purpose of src and alt in an image?
10. Differentiate between absolute and relative image paths.
11. What is the purpose of href?
12. Differentiate between ordered and unordered lists.
13. Explain nested lists with an example.
14. What are rowspan and colspan?
15. Explain form, input, label, and button.
16. Differentiate between radio buttons and checkboxes.
17. What is the purpose of name in a form control?
18. Differentiate between block-level and inline elements.
19. Explain the difference between div and span.
20. What is the difference between GET and POST form methods?

## B. Practical assignments

## Lab Task 1: Personal introduction page

- Your name as the main heading.
- A paragraph introducing yourself.
- At least three subheadings.
- Bold, italic, and emphasized text.
- A horizontal rule separating two sections.

## Lab Task 2: Images and hyperlinks

- Three images using relative paths and meaningful alt text.
- An external website link and a link that opens in a new tab.

- An in-page link to a contact section.
- An email link.

## Lab Task 3: Lists

- An ordered list of five places to visit.
- An unordered list of five hobbies.
- A nested list of frontend and backend technologies.
- A description list explaining HTML, CSS, and JavaScript.

## Lab Task 4: Student timetable

- Days of the week, period timings, and subject names.
- A break row using colspan.
- A suitable table caption.

## Lab Task 5: Student registration form

- Full name, email, password, and date of birth.
- Gender using radio buttons and skills using checkboxes.
- Course dropdown and address textarea.
- Submit and reset buttons, labels, and required fields.

## Lab Task 6: Block and inline elements

- Two separate div containers with paragraphs.
- A span around selected words.
- Links and bold text inside a paragraph.
- Use class and id attributes.

# Quick Revision

| Concept | Main tag or syntax |
| --- | --- |
| HTML document | <!DOCTYPE html>, html |
| Metadata | head, meta, title |
| Visible content | body |
| Headings | h1 to h6 |
| Paragraph | p |
| Text formatting | strong, em, b, i, mark |
| Line break / thematic break | br / hr |
| Image / hyperlink | img / a |
| Ordered / unordered list | ol, li / ul, li |
| Description list | dl, dt, dd |
| Table | table, tr, th, td |
| Form controls | form, label, input, button |
| Multiline input / dropdown | textarea / select, option |
| Grouped form controls | fieldset, legend |
| Generic block / inline container | div / span |

Final tip: Practice every example in an index.html file and open it in a browser. Change the content and attributes, then observe how the output changes.